



Politecnico
di Bari

DIPARTIMENTO
INTERATENEO
DI FISICA



Dottorato in Fisica – XL ciclo - 2025

Machine Learning techniques for particle physics

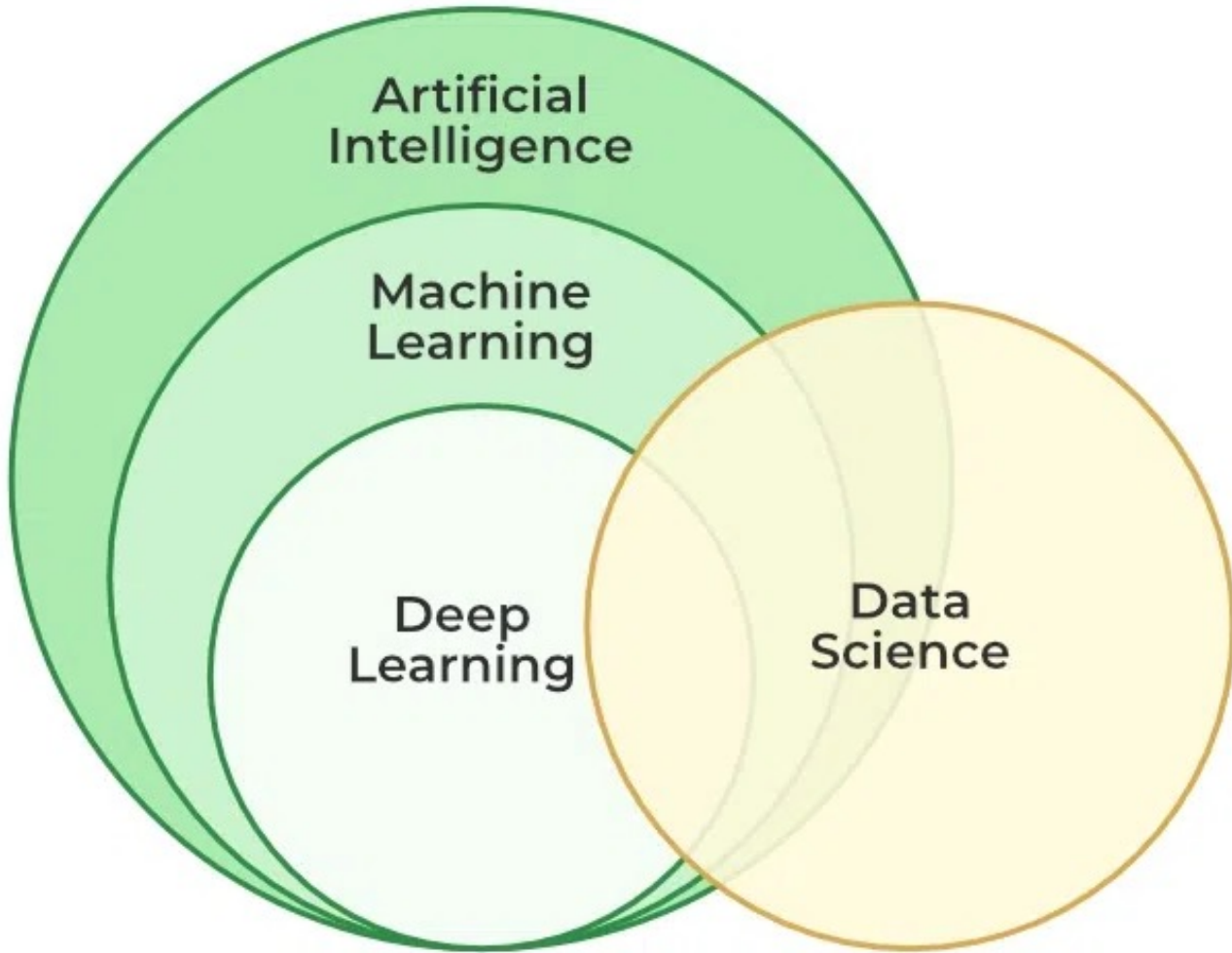
Federica Maria Simone (Poliba & INFN) - federica.simone@poliba.it

Adriano Di Florio (CC-IN2P3) - adriano.di.florio@cern.ch

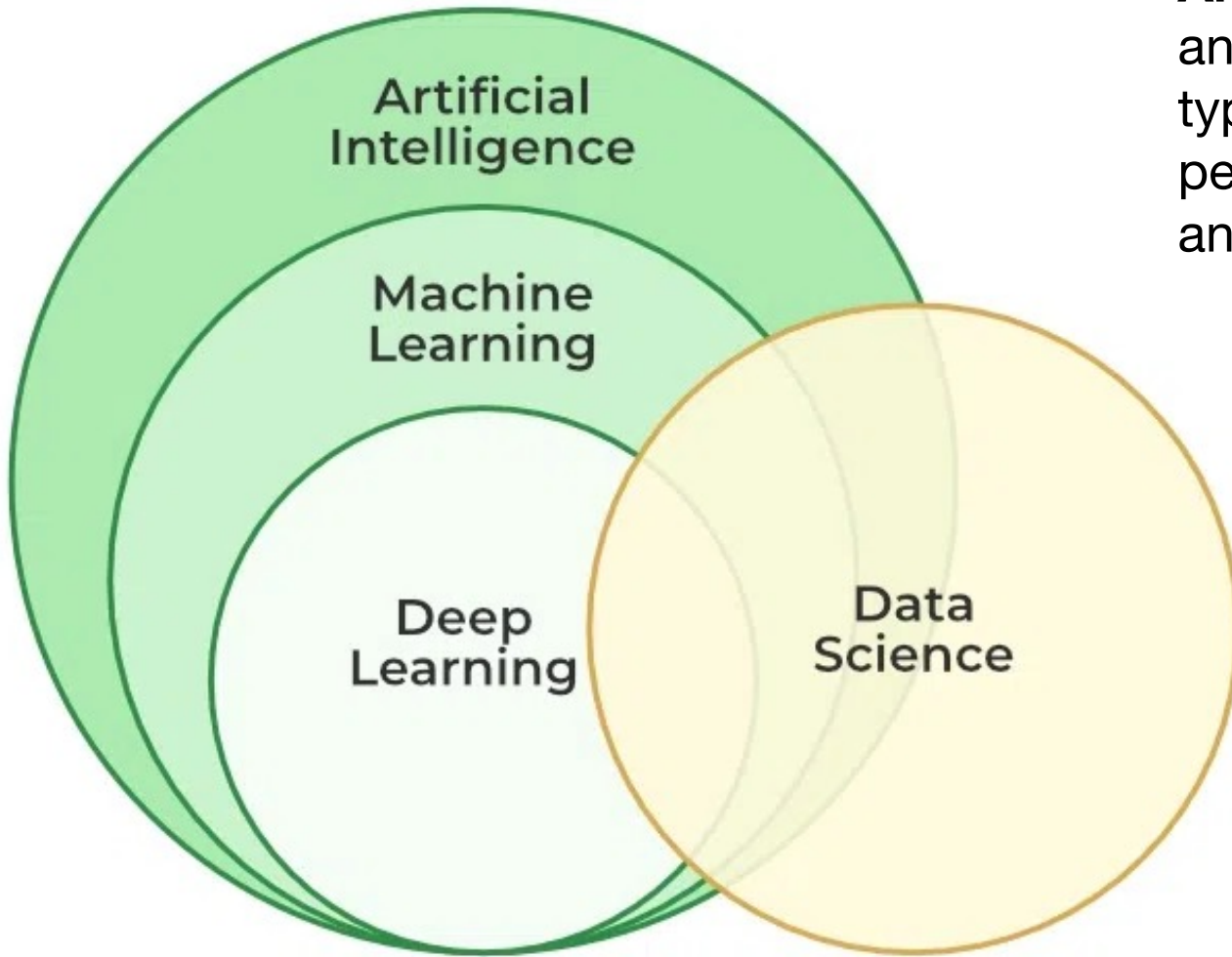
What is Machine Learning

Disclaimer: these slides are taken from many sources, please find useful links and credits on my last slide!

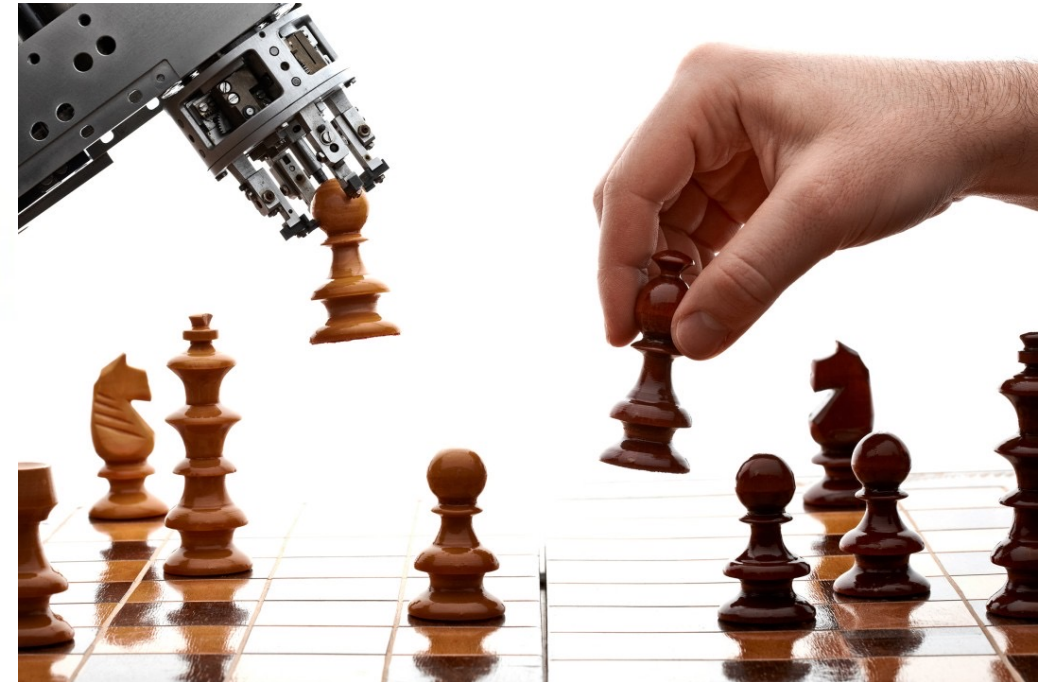
What is Machine Learning



What is Machine Learning

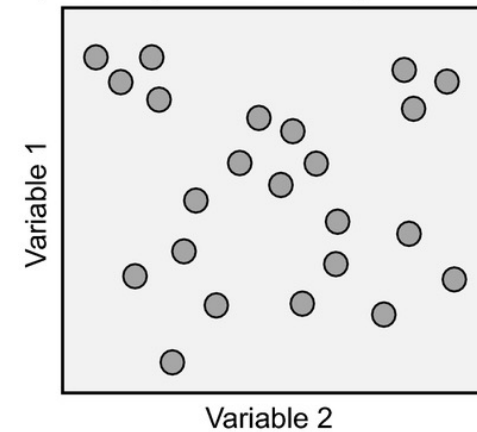
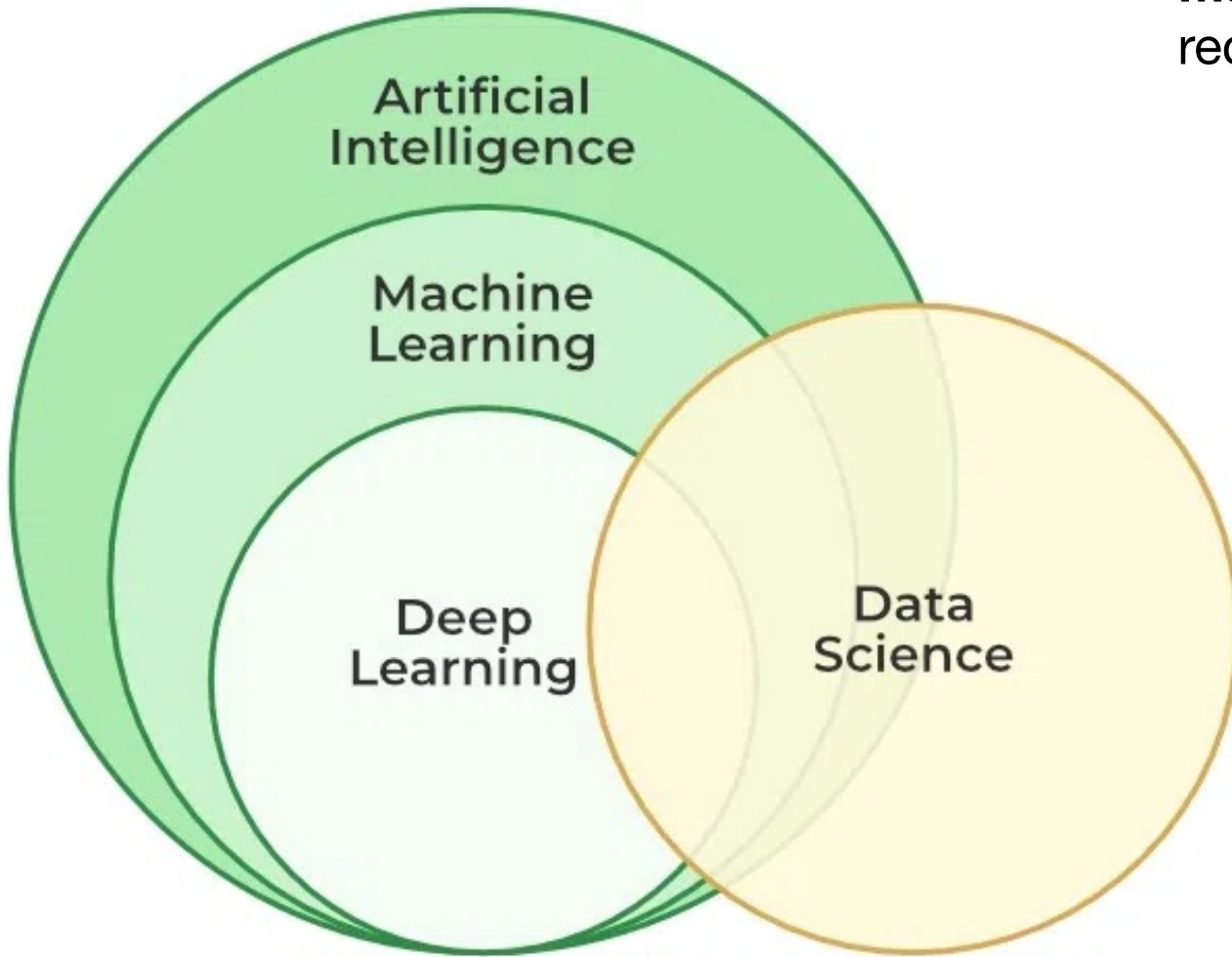


- **Artificial Intelligence:** development of algorithms and computer programs that can perform tasks that typically require human intelligence such as visual perception, speech recognition, decision-making, and language translation



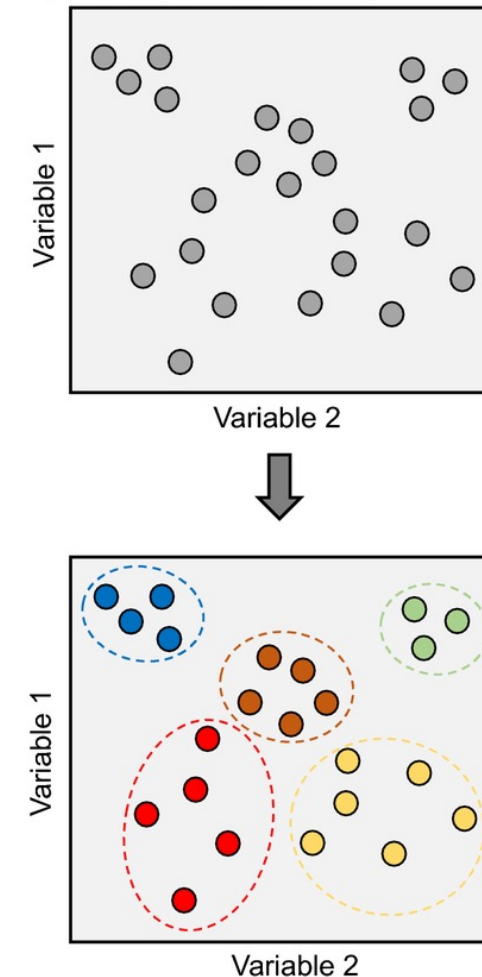
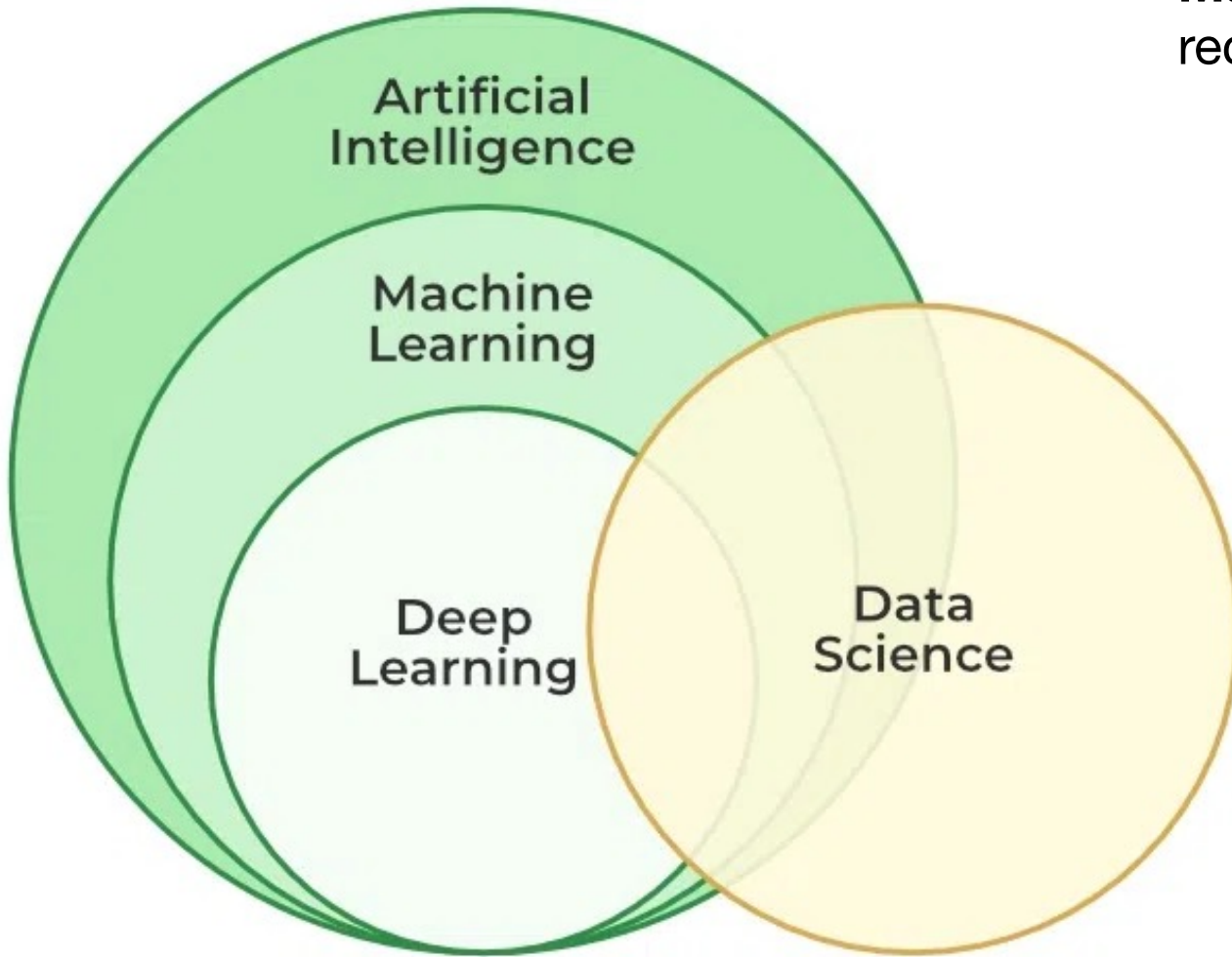
What is Machine Learning

- **Machine Learning:** automatically learn insights and recognize patterns from data.



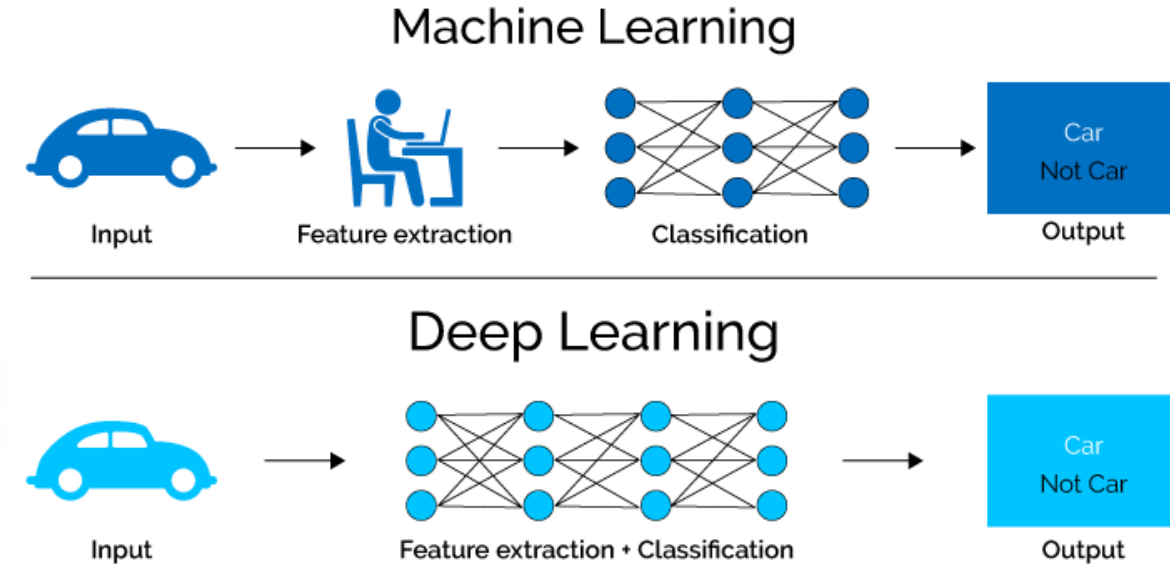
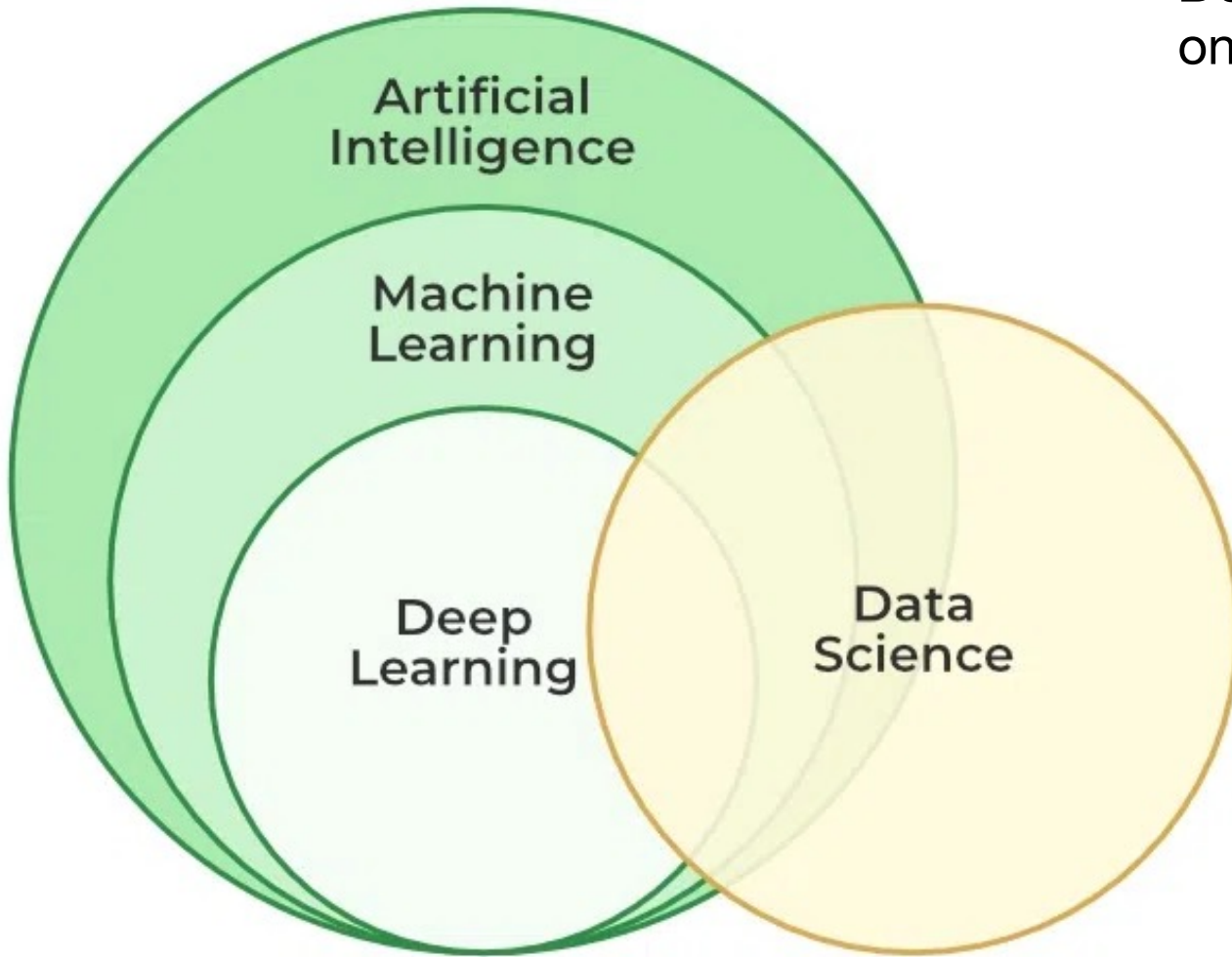
What is Machine Learning

- **Machine Learning:** automatically learn insights and recognize patterns from data.



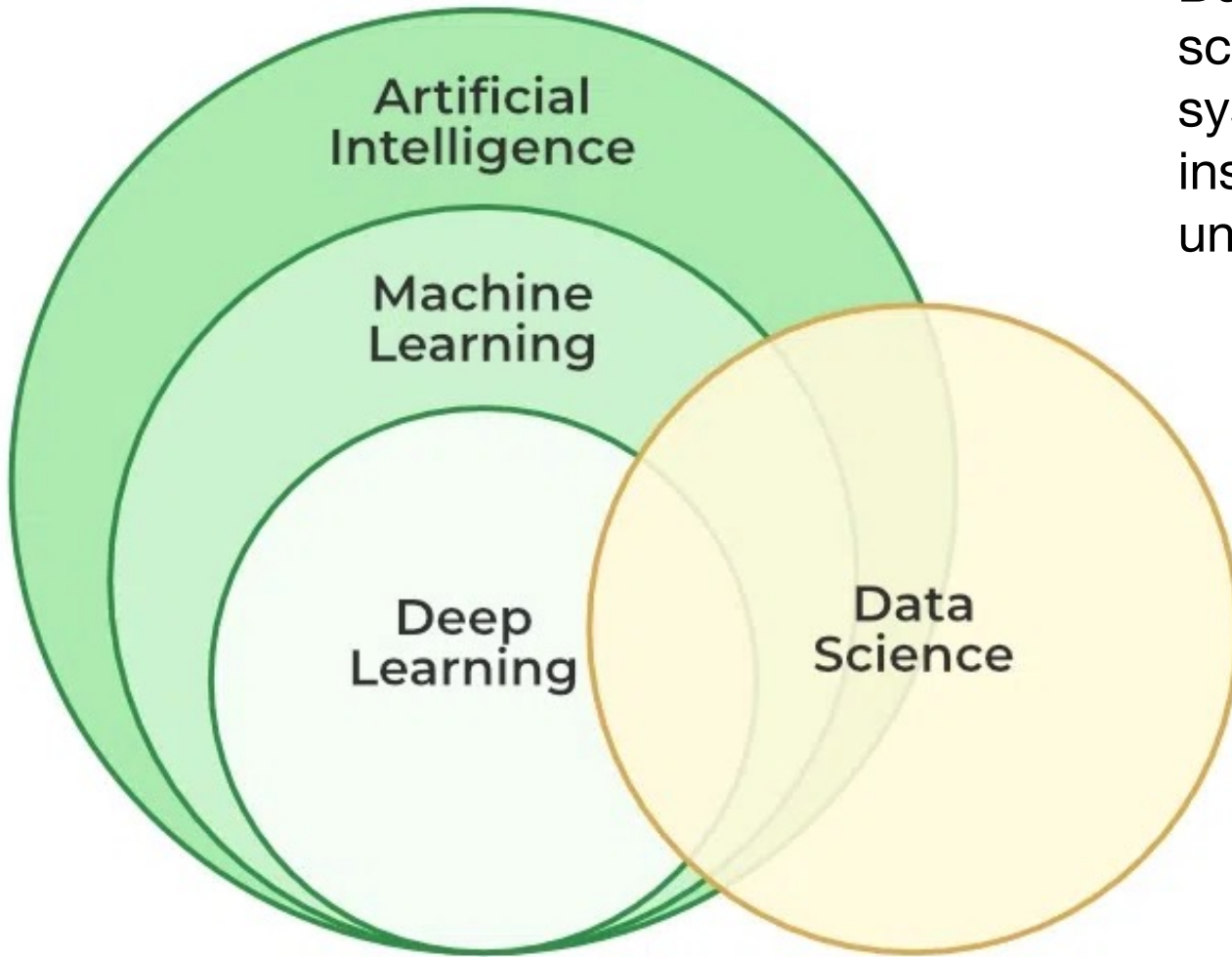
What is Machine Learning

- **Deep Learning:** complex neural networks trained on large amount of unstructured data.



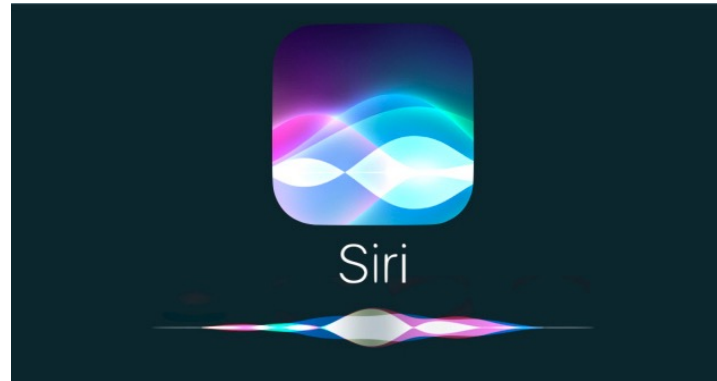
What is Machine Learning

- **Data Science:** uses statistics, scientific computing, scientific methods, processes, algorithms and systems to extract or extrapolate knowledge and insights from potentially noisy, structured, or unstructured data.

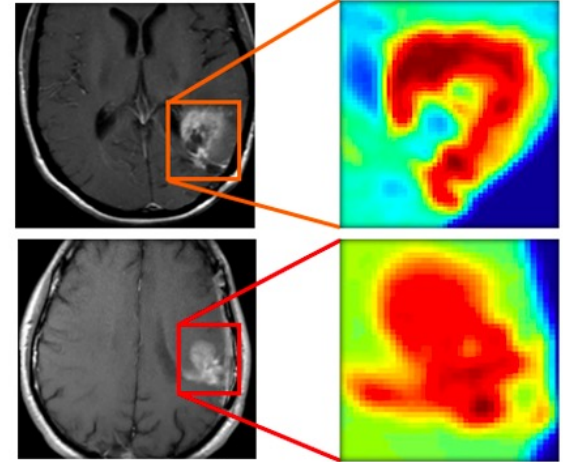


Some applications ...

- Natural Language Processing
- Speech and handwriting recognition
- Object recognition and computer vision
- Fraud detection
- Financial market analysis
- Search engines
- Spam and virus detection
- Medical diagnosis
- Robotics control
- Automation: energy usage, systems control, video games, self-driving cars
- Advertising
- Data Science

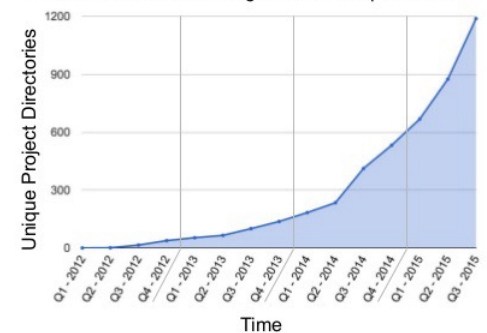


DIAGNOSTICS



Growing Use of Deep Learning at Google

of directories containing model description files



Across many products/areas:

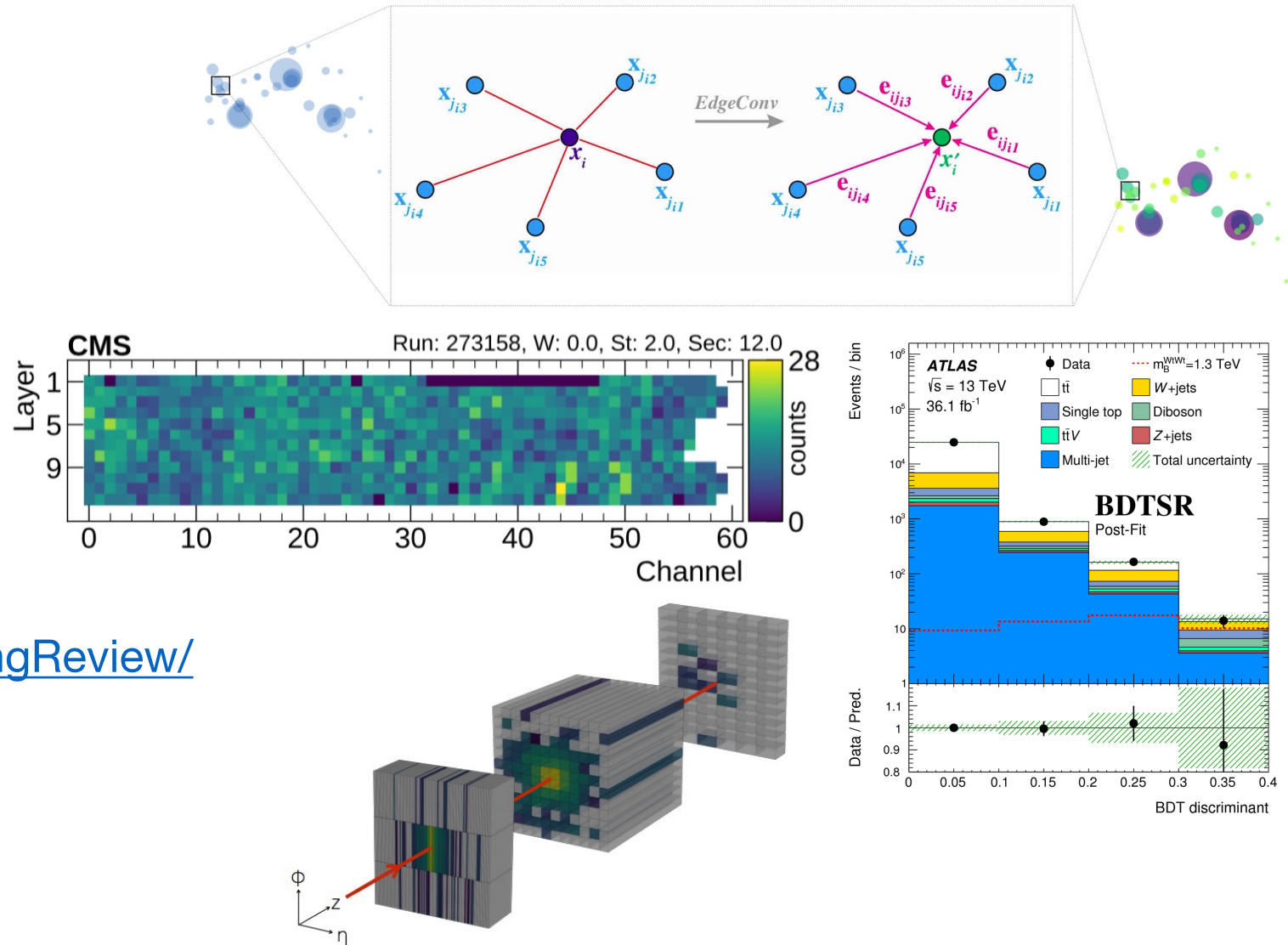
Android
Apps
drug discovery
Gmail
Image understanding
Maps
Natural language understanding
Photos
Robotics research
Speech
Translation
YouTube
... many others ...



... in HEP

- Particle tagging
 - Signal classification
 - Anomaly detection
 - Fast simulation
 - Simulation-based inference
 - Design optimisation
 - Uncertainty mitigation
- ... and more

<https://iml-wg.github.io/HEPML-LivingReview/>



A rapidly evolving field

The ChatGPT logo, featuring the text "ChatGPT" in white on a background of vertical purple and black stripes.

Stable Diffusion



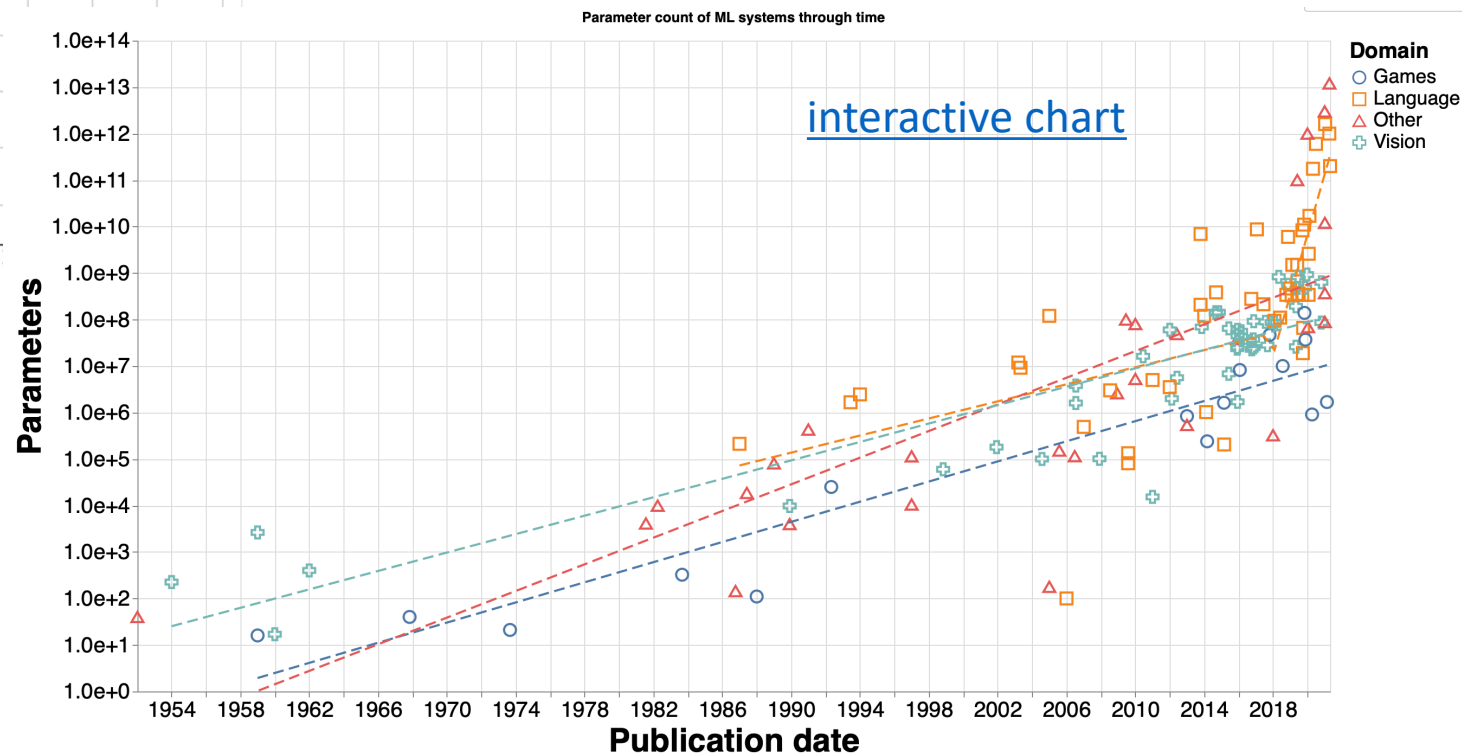
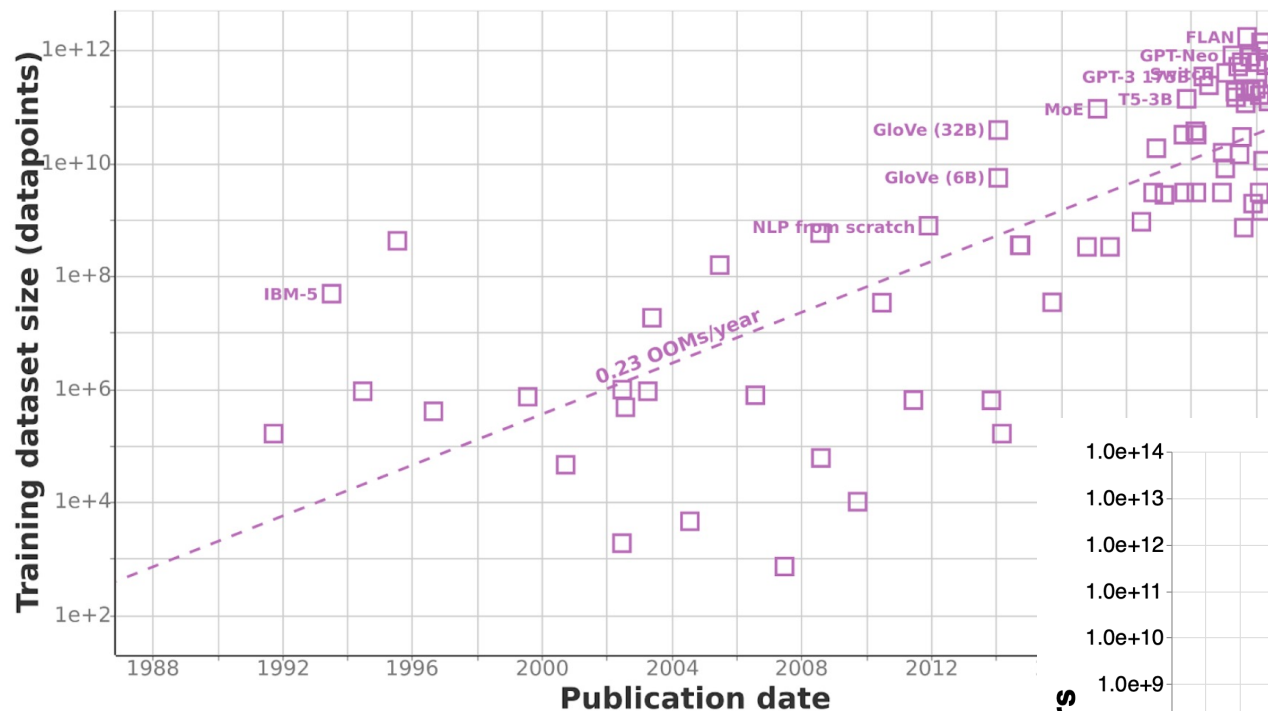
GitHub Copilot

Prompt:

*street style photo of a woman
selling pho at a Vietnamese
street market, sunset,
shot on fujifilm*



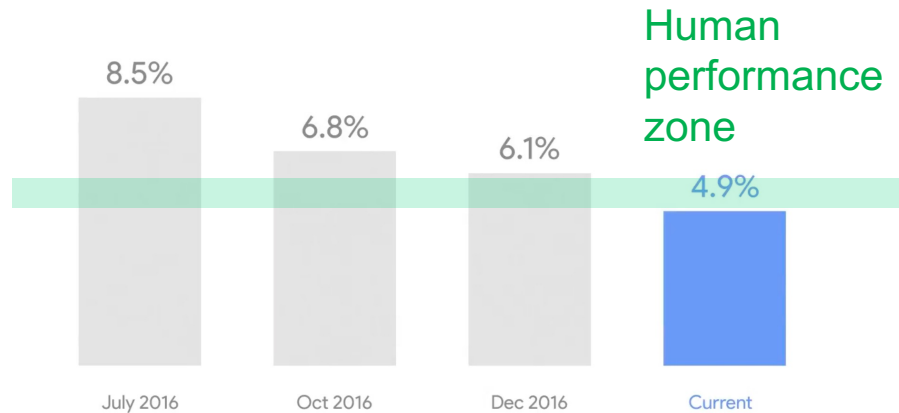
Increasing sizes and complexity ...



... and performance

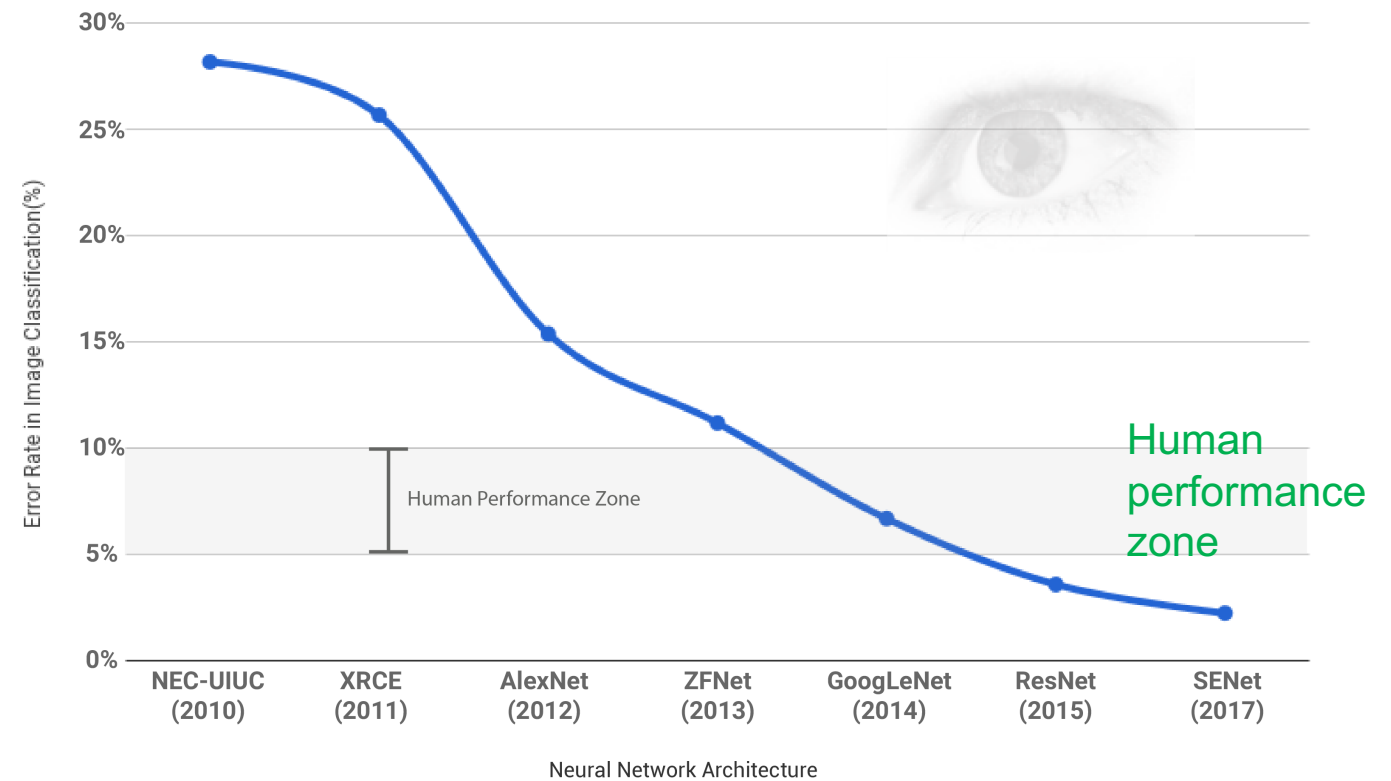
Speech Recognition

Word Error Rate



2017 Google results

US English only.



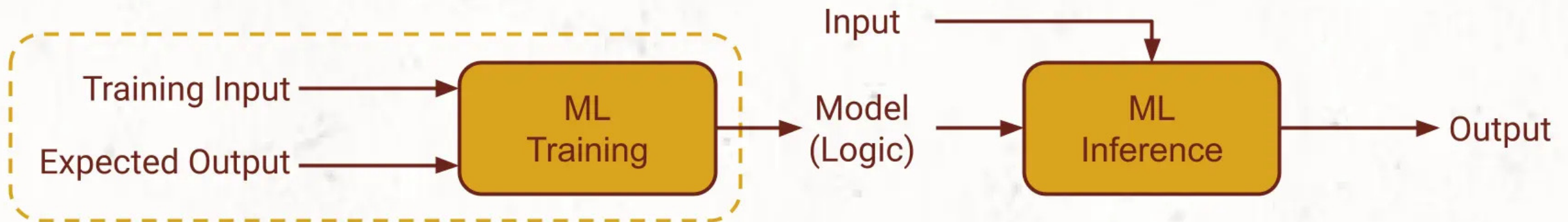
<https://arxiv.org/pdf/1409.0575.pdf>

Let's start from basics: learning algorithms

Traditional Programs: Define algo/logic to compute output

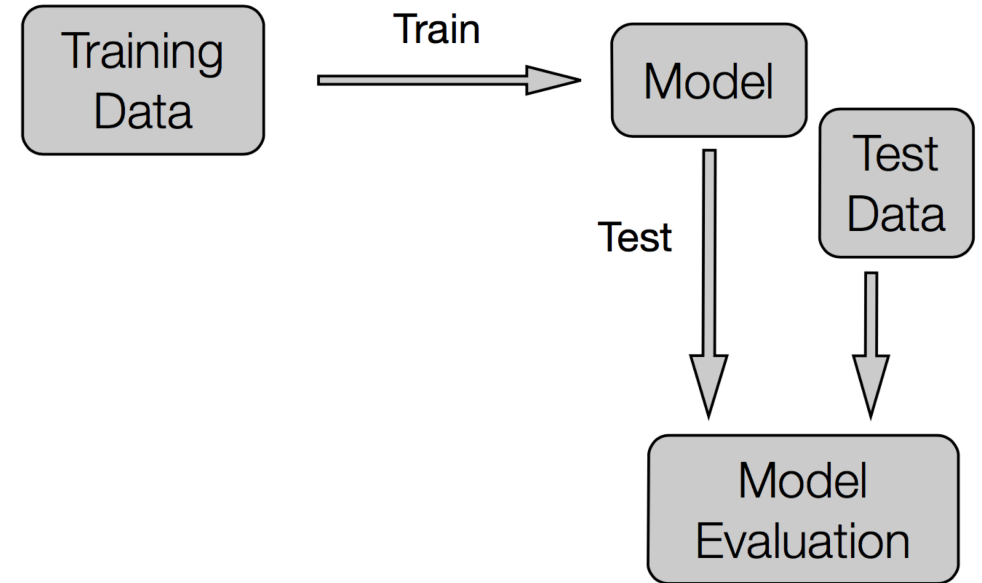


Machine Learning: Learn model/logic from data



Machine learning models

- Key element in machine learning is a **mathematical model**
- A mathematical characterization of system(s) of interest, typically via **random variables**
- Chosen model **depends on the task** / available data
- **Learning**: estimate statistical model from data
- **Prediction and Inference**: using statistical model to make predictions on new data points and infer properties of system(s)



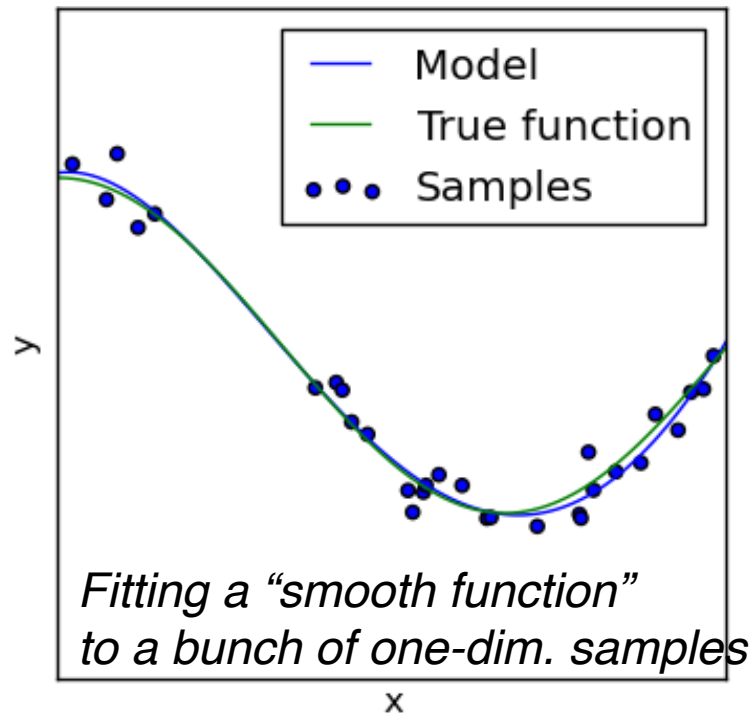
Parametric vs nonparametric models

Parametric: “fixed-size” models that do not “grow” with the data

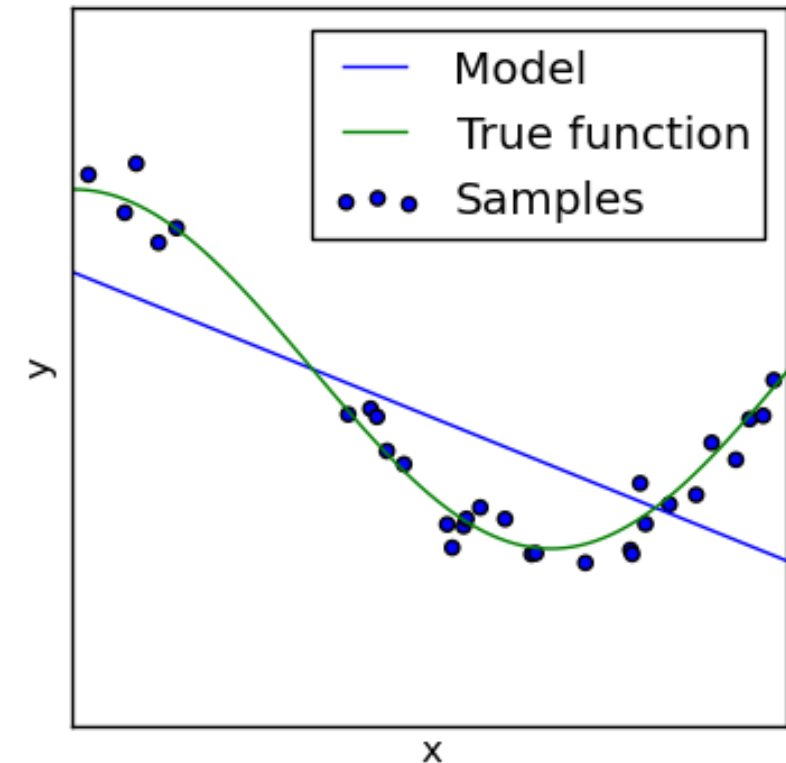
- More data just means you learn/fit the model better

Nonparametric: models that grow with the data

- More data means a more complex model

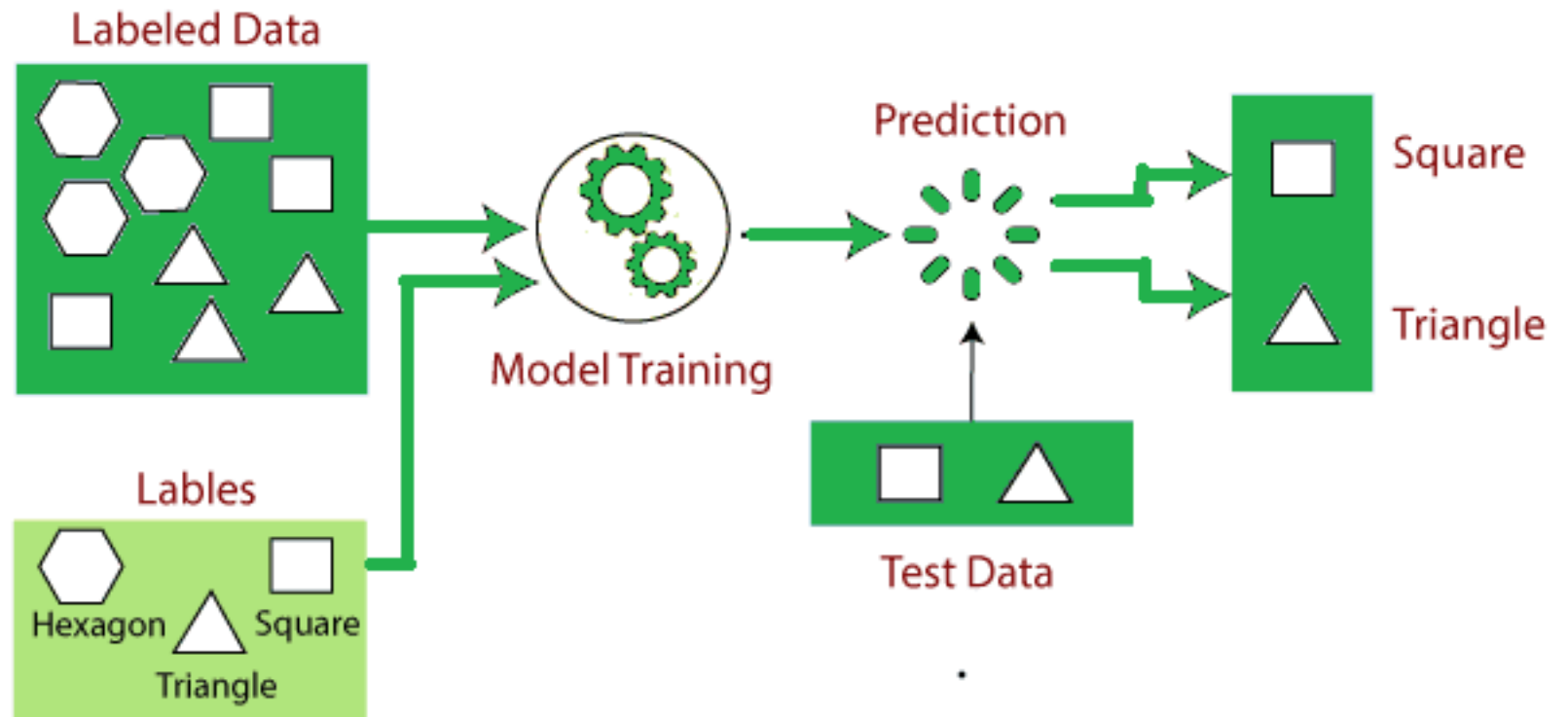


Fitting a simple line (2 params) to a bunch of one-dim. samples



How: supervised learning

- The desired output (target) is known at training time.
 - calculate an error based on target output and algorithm output
 - use error to make corrections by updating the weights.
- Given N examples with features $\{x_i\}$ and targets $\{y_i\}$, learn function mapping $\mathbf{h(x)=y}$

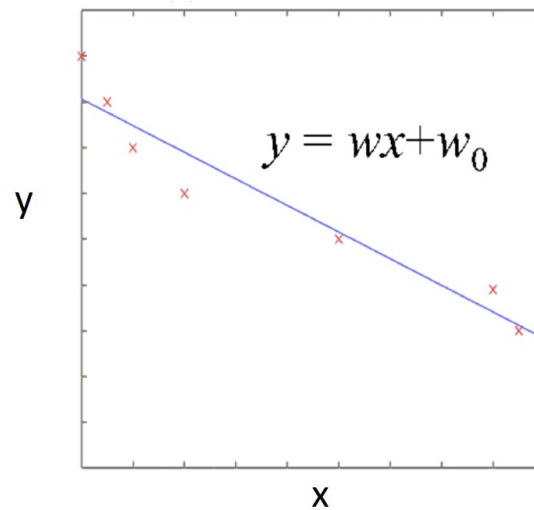


How: supervised learning

- The desired output (**target**) is known at training time.
 - calculate an **error** based on **target output** and **algorithm output**
 - use error to make corrections by **updating the model parameters (weights)**.
- Given N examples with features $\{x_i\}$ and targets $\{y_i\}$, learn function mapping **$h(x)=y$**

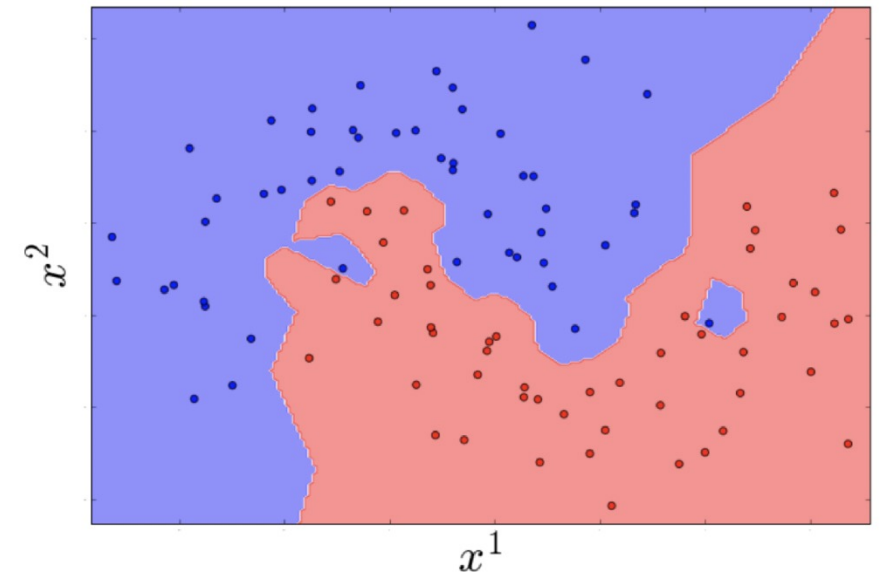
- **Regression:**

- $Y = \text{Real Numbers}$



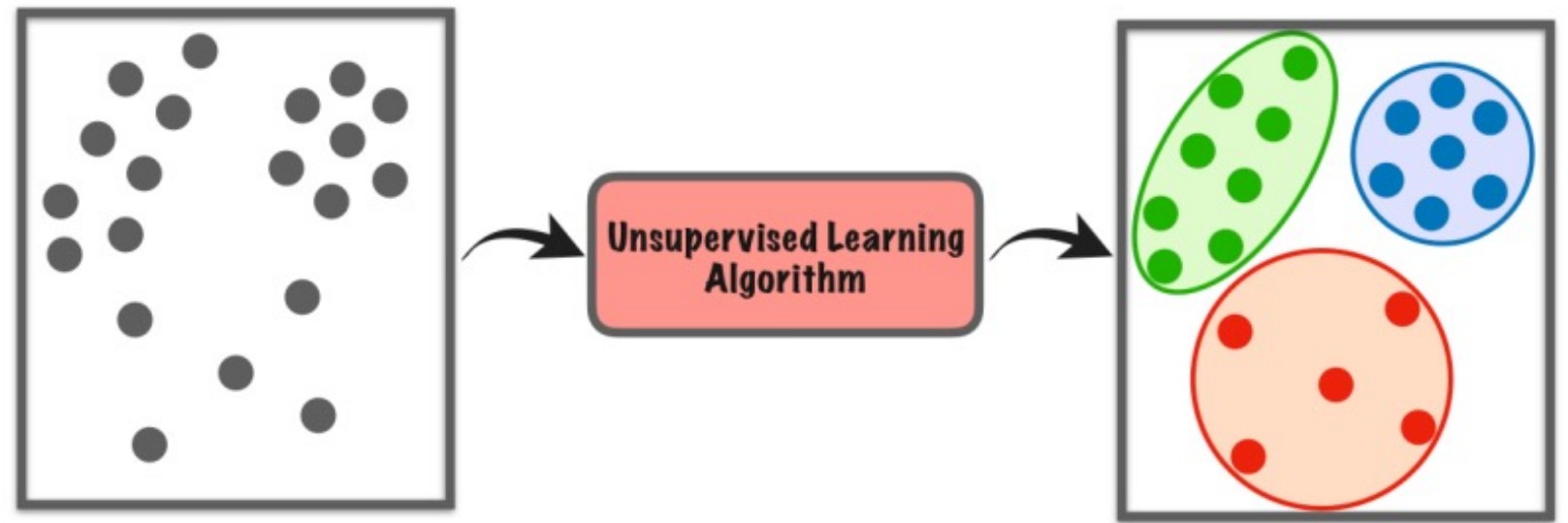
- **Classification:**

- Y is a finite set of labels



How: unsupervised learning

- No target output is used at training time
 - The ML algorithm finds pattern within the inputs
- Given some data $D=\{x_i\}$, but no labels $\rightarrow$ find structure in the data



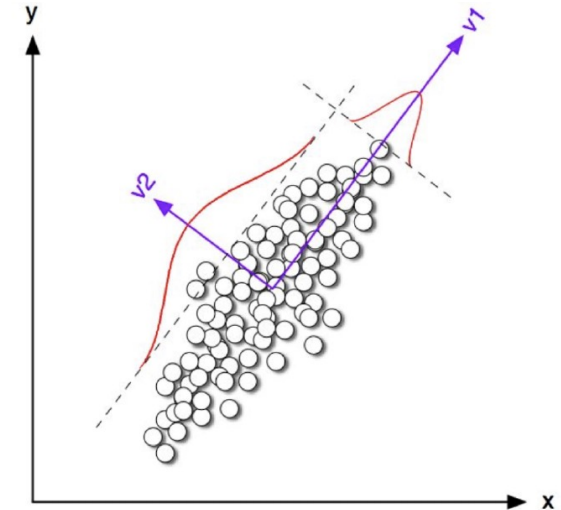
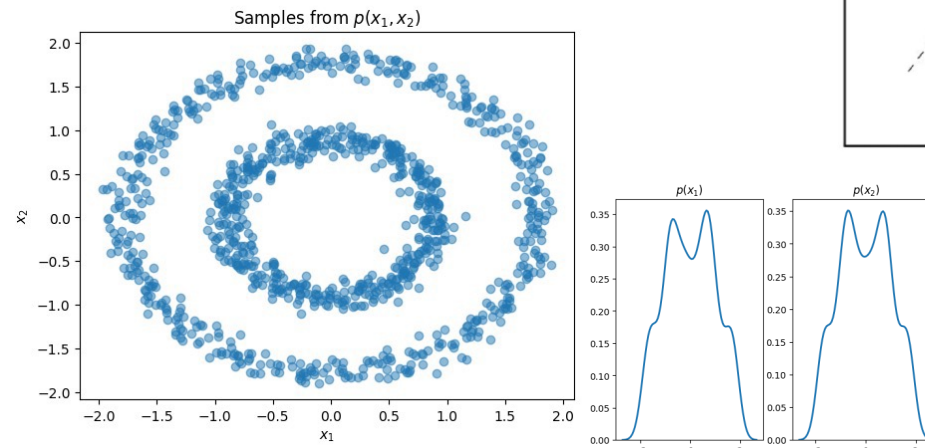
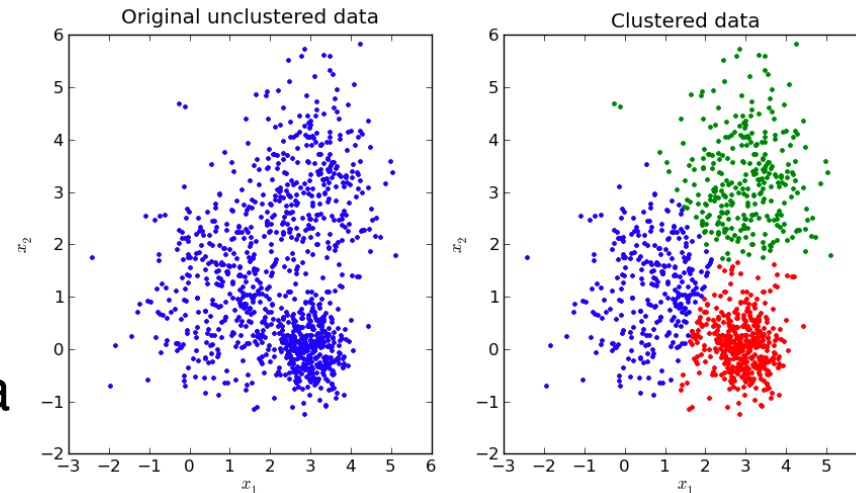
How: unsupervised learning

- No target output is used at training time
 - The ML algorithm finds pattern within the inputs
- Given some data $D=\{x_i\}$, but no labels $\rightarrow$ find structure in the data

- **Clustering:**
partition the data into groups

- **Dimensionality reduction:** find a low dimensional (less complex) representation of the data with a mapping $Z=h(X)$

- **Density estimation and sampling:** estimate the PDF $p(x)$, and/or learn to draw plausible new samples of x



Terminology: ML

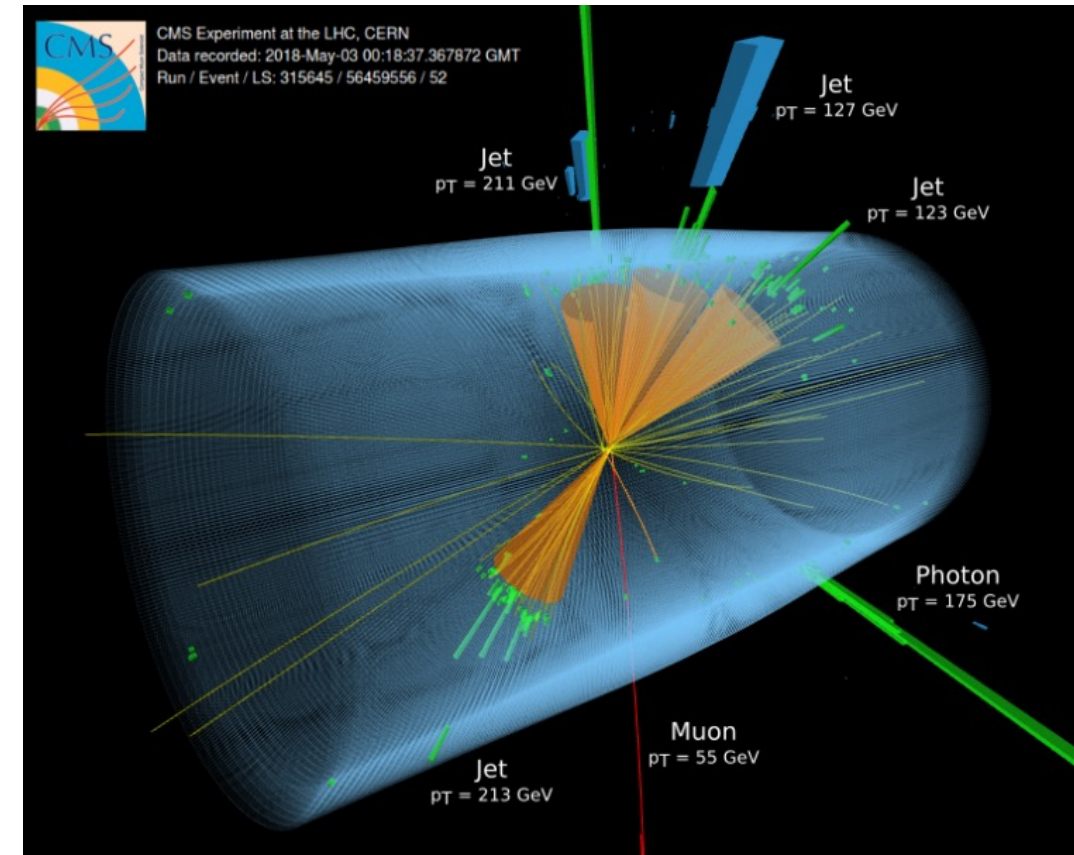
1. **Instance/example:** The values of one row of features and possibly a label.
2. **Dataset:** A collection of instances
3. **Feature:** Input variable to your ML model

Features			Label
Temperature	Humidity	Pressure	Test score
15	47	998	Good
19	34	1020	Excellent
18	92	1012	Poor

Terminology: HEP

(High Energy) Particle Physics

1. **Event:** All information collected during a collision inside a detector, or reproduced from a Monte Carlo simulation of such collisions (**instance/example**)
2. **Sample:** A collection of events, a **dataset**.
3. **Variable:** A property of the event or of one of its constituents (**feature**)
4. **Cut:** To cut on a variable is to apply a threshold on this variable and keep only events satisfying this condition. A cut-based analysis is applying such thresholds on several variables to select events.
5. **Event weight:** In high-energy physics events usually have an associated weight, which depends on how many events were generated (relating to the process cross section and collected luminosity) and various corrections applied to simulations to account for differences between data and Monte Carlo predictions (jet energy scale or object identification efficiency are such weights)



References and useful links

- <http://scikit-learn.org/>
- Pattern Recognition and Machine Learning, Bishop (2006)
- Elements of Statistical Learning (2nd Ed.) Hastie, Tibshirani & Friedman 2009
- Introduction to machine learning, Murray http://videlectures.net/bootcamp2010_murray_iml/
- What is Machine Learning, Ravikumar and Stone
http://www.cs.utexas.edu/sites/default/files/legacy_files/research/documents/MLSSIntro.pdf
- CS181, Parkes and Rush, Harvard University <http://cs181.fas.harvard.edu>
- CS229, Ng, Stanford University <http://cs229.stanford.edu/>
- François Fleuret's deep-learning courses 14x050 of the University of Geneva,
<https://fleuret.org/dlc/#lectures>
- Machine learning in high energy physics, Alex Rogozhnikov <https://indico.cern.ch/event/497368/>
- Introduction to Machine Learning and Deep Learning, Micheal Kagan
<https://indico.cern.ch/event/1293858/>
- Public HEP datasets: <https://iml.web.cern.ch/public-datasets>
- A Living Review of Machine Learning for Particle Physics: <https://iml-wg.github.io/HEPML-LivingReview/>
- Machine Learning in High Energy Physics Community White Paper <https://arxiv.org/abs/1807.02876>

Python ML software

- Anaconda / Conda → easy to setup python ML / scientific computing environments
 - <https://www.continuum.io/downloads>
 - <http://conda.pydata.org/docs/get-started.html>
- Integrating ROOT / PyROOT into conda
 - <https://nlesc.gitbooks.io/cern-root-conda-recipes/content/index.html>
 - <https://conda.anaconda.org/NLeSC>
- Converting ROOT trees to python numpy arrays / panda dataframes
 - https://pypi.python.org/pypi/root_numpy/
 - https://github.com/ibab/root_pandas
- Scikit-learn → general ML library
 - <http://scikit-learn.org/stable/>
- Deep learning frameworks / auto-differentiation packages
 - <https://www.tensorflow.org/>
 - <http://deeplearning.net/software/theano/>
- High level deep learning package build on top of Theano / Tensorflow
 - <https://keras.io/>